

Commerce XI Mathematics & Statistics · Part 1

Unit 2: Functions

Symbols, Laws, Theorems, Results and Formulae

A quick reference based on Unit 2, textbook pages 20–32.

How to use this guide

Use	What to check
Before solving	Identify the function, its domain and co-domain, and the operation or value required.
During solving	Use the correct rule, preserve every domain restriction, and apply compositions from right to left.
Before answering	Check the required notation, endpoints, excluded values, and whether the stated range is correct.

Scope

Function notation · domain, co-domain and range · standard functions · graphs · algebra of functions · composition · inverse functions · piecewise functions.

1. Function notation and symbols

Symbol	How to read it	Meaning
$f: A \rightarrow B$	f from A to B	A is the domain and B is the co-domain.
$y = f(x)$	y equals f of x	x is a pre-image and y is its unique image.
$f(A)$	image of A under f	The range of f ; it is a subset of the co-domain.
$g \circ f$	g composed with f	Apply f first and then apply g .
f^{-1}	inverse of f	Exists as a function when f is bijective.
$\mathbb{R} - \{0\}$	real numbers except zero	Common domain or range restriction for reciprocal functions.
$[a, b]$	closed interval a to b	Both endpoints are included.
(a, b)	open interval a to b	Both endpoints are excluded.

2. Core definitions

Term	Definition / test
Function	Every element of the domain has exactly one image in the co-domain.
Domain	The complete set of permitted input values.
Co-domain	The declared destination set of a function.
Range	The set of values actually obtained: $\{f(x) / x \text{ belongs to the domain}\}$.
One-one (injective)	Distinct inputs have distinct images. Equivalently, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
Onto (surjective)	Every element of the co-domain has at least one pre-image; range = co-domain.
Into	At least one element of the co-domain has no pre-image; range is a proper subset of the co-domain.
Bijective	The function is both one-one and onto.

3. Standard functions

Function	Domain	Range	Key graph fact
Constant: $f(x) = k$	$\mathbb{R}$	$\{k\}$	Horizontal line $y = k$.
Identity: $f(x) = x$	$\mathbb{R}$	$\mathbb{R}$	Straight line through the origin with slope 1.
Linear: $f(x) = ax + b, a \neq 0$	$\mathbb{R}$	$\mathbb{R}$	Slope a and y -intercept b .
Quadratic: $f(x) = ax^2 + bx + c$	$\mathbb{R}$	Depends on its maximum or minimum	Parabola; opens upward for $a > 0$ and downward for $a < 0$.
Square: $f(x) = x^2$	$\mathbb{R}$	$[0, \infty)$	Even function; symmetric about the y -axis.
Cube: $f(x) = x^3$	$\mathbb{R}$	$\mathbb{R}$	Odd function; passes through the origin.
Reciprocal: $f(x) = 1/x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$	The axes are asymptotes.
Exponential: $f(x) = a^x; a > 0, a \neq 1$	$\mathbb{R}$	$(0, \infty)$	Passes through $(0, 1)$.
Logarithmic: $f(x) = \log_a x$	$(0, \infty)$	$\mathbb{R}$	Inverse of $y = a^x$; passes through $(1, 0)$.
Signum: $\text{sgn}(x)$	$\mathbb{R}$	$\{-1, 0, 1\}$	Values depend on whether x is negative, zero or positive.
Absolute value: $f(x) = x $	$\mathbb{R}$	$[0, \infty)$	V-shaped graph; an even function.
Greatest integer: $f(x) = [x]$	$\mathbb{R}$	$\mathbb{I}$	Step graph; $[x]$ is the greatest integer $\leq x$.

4. Algebra of functions

Operation	Formula	Domain condition
Sum	$(f + g)(x) = f(x) + g(x)$	Common domain of f and g .
Difference	$(f - g)(x) = f(x) - g(x)$	Common domain of f and g .
Product	$(fg)(x) = f(x)g(x)$	Common domain of f and g .
Quotient	$(f/g)(x) = f(x)/g(x)$	Common domain, with $g(x) \neq 0$.
Composition	$(g \circ f)(x) = g[f(x)]$	x must be in the domain of f , and $f(x)$ must be in the domain of g .

Order matters in composition

In general, $f \circ g$ and $g \circ f$ need not be equal. Always substitute the inner function first and retain every domain restriction introduced by the inner and outer functions.

5. Inverse functions

Result	Meaning / method
f^{-1} exists	For this unit, verify that f is one-one and onto (bijective).
$f^{-1}[f(x)] = x$	The inverse reverses the action of f .
$f[f^{-1}(x)] = x$	The other composite is also the identity function.
Domain of f^{-1}	Range of f .
Range of f^{-1}	Domain of f .
Finding an inverse	Write $y = f(x)$, solve for x in terms of y , interchange x and y , and state the correct domain.

6. Evaluating and reading functions

Situation	Method
Formula	Replace the independent variable by the given input and simplify carefully.
Ordered pairs	The first components form the domain; the second components form the range.
Arrow diagram	Each domain element must have exactly one outgoing arrow for the relation to be a function.
Graph	To find $f(a)$, locate $x = a$ and read the corresponding y-coordinate.
Piecewise function	First decide which condition contains the input, then use only the corresponding rule. Check endpoint symbols $\leq$, $<$, $\geq$ and $>$.

7. Important results and checks

Result / check	What to remember
Function test	No domain element may be missing an image or have more than one image.
Range versus co-domain	They are equal only for an onto function.
Denominator	Exclude every input that makes a denominator zero.
Logarithm	Its argument must be positive; the base must satisfy $a > 0$ and $a \neq 1$.
Square root	For real-valued functions, the expression under an even root must be non-negative.
Piecewise endpoints	Use the branch whose inequality actually includes the endpoint.
Composition	Work from right to left: $(g \circ f)(x)$ means $g[f(x)]$.
Inverse	Do not confuse $f^{-1}(x)$ with $1/f(x)$.

Use this guide for revision before attempting Unit 2 tests. For worked calculations, submit a paper and download its Detailed Answers PDF from the result page.